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The Integration of Artificial Intelligence in Search Engine Optimization: The benefits and the challenges for marketers

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Master Thesis

presented as partial requirement for obtaining a Master's Degree in Data-Driven Marketing

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The Integration of Artificial Intelligence in Search Engine Optimization: The benefits and the challenges for marketers

by

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Master Thesis presented as partial requirement for obtaining the Master's degree in Data-Driven Marketing, with a specialization in Digital Marketing and Analytics

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STATEMENT OF INTEGRITY

I hereby declare having conducted this academic work with integrity. I confirm that I have not used plagiarism, any form of undue use of information or falsification of results along the process leading to its elaboration. I further declare that I have fully acknowledged the Rules of Conduct and Code of Honor from the NOVA Information Management School.

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ABSTRACT

In recent years, the digital environment has experienced significant transformation, largely driven by the increasing influence of artificial intelligence (AI) in shaping search engine optimization (SEO) strategies and transforming the role of marketing professionals. This study investigates the impact of AI integration in SEO by comparing AI-driven and human-driven approaches through an experimental research design. The findings indicate that while AI-based SEO significantly enhances marketers' perceived productivity, it does not improve perceived efficiency. Mediation analyses revealed that neither ethical concerns nor fear of replacement significantly accounted for this relationship. Similarly, moderation analysis showed that the level of AI knowledge did not significantly affect the influence of AI integration on productivity or efficiency. These results suggest that although AI contributes positively to output quality, its effects on perceived effort reduction and psychological drivers may be more limited than previously assumed. The findings highlight important theoretical and managerial implications, emphasizing that the benefits of AI in SEO depend not only on its capabilities, but also on users' perceptions, trust, and practical integration into existing workflows. The study offers new insights into the complex and evolving role of AI in digital marketing workflows, with implications for both theory and practice.

KEYWORDS

Search Engine Optimization; Artificial Intelligence; Productivity; Efficiency; Ethical Concerns; Fear of Replacement

Sustainable Development Goals (SDG):

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LIST OF ABBREVIATIONS AND ACRONYMS

AI	Artificial Intelligence
SEO	Search Engine Optimization
DV	Dependent Variable
IV	Independent Variable
B2B	Business-to-Business
NLP	Natural Language Processing
AIO	Artificial Intelligence Optimization
GEO	Generative Engine Optimization
AEO	Answer Engine Optimization
E-E-A-T	Experience, Expertise, Authoritativeness, Trustworthiness
ML	Machine Learning
AWS	Amazon Web Services
RNN	Random Neural Networks

1. INTRODUCTION

In today's digital era, search engines have become the primary tool through which users access information, products, and services. Their central role in the online experience makes visibility within search results a critical factor for business success. According to Semrush, a leading digital marketing and Search Engine Optimization (SEO) analytics platform, Google processes approximately 5.9 million searches every minute, translating into nearly 8.5 billion searches per day (Padrig Jones, 2024). This exceptionally high level of search activity highlights how frequently and heavily users depend on search engines, while also revealing the immense competition for visibility in the digital space. As a result, Search Engine Optimization (SEO) has evolved into a fundamental component of digital marketing strategies, directly influencing brand visibility, website traffic, and consumer engagement.

Artificial intelligence (AI) has radically changed digital marketing, modifying traditional methods and creating opportunities for innovative strategies that improve efficiency, personalization and engagement. AI, once considered an innovative concept, has now entered marketers' everyday lives, influencing their habits and expectations. Recent studies confirm this trend: (IBM, 2022) surveyed 7,502 respondents worldwide and found that 35 percent of firms used AI, while a 2023 McKinsey & Company report indicated that one-third of organizations regularly apply AI technologies in their operations (McKinsey, 2023). In the marketing industry, AI-based technologies, enhanced by big data, advanced algorithms and powerful computing systems, have become necessary tools for marketers (Davenport et al., 2020; Overgoor et al., 2019).

Among the various uses of AI in digital marketing, its integration into SEO is a highly transformative development by enhancing website visibility, content optimization, and search rankings (Goodwin, 2023; Yalçın & Köse, 2010). Once focused on keywords and backlinks, SEO now prioritizes content quality, user experience, and technical optimization (Matta et al., 2020). AI-driven tools like Google's RankBrain and BERT use machine learning and natural language processing to analyze user behavior, predict search trends, and optimize content dynamically (Akay et al., 2018; Go, 2024).

These advancements allow marketers to decode search algorithms, refine keyword targeting, and enhance on-page and off-page SEO. AI also improves voice search and image optimization, enhancing user interactions and accessibility (Amara et al., 2024; Ziakis & Vlachopoulou, 2024). With 61% of marketers prioritizing SEO for organic growth, AI-driven SEO remains a key strategy for boosting brand awareness, attracting traffic, and increasing conversions (Cushman, 2018; Southern, 2024; *The Ultimate List of Marketing Statistics for 2024*, 2024) Beyond these technical enhancements, AI also delivers measurable improvements in marketer productivity and efficiency. Studies show that 80% of employees using AI tools report increased productivity, with professionals saving up to 1.75 hours per day—time that can be redirected toward higher-value marketing tasks. Furthermore, over 54% of businesses surveyed by IBM reported cost savings and notable productivity gains following the adoption of AI in their operations (Gustav Westman, 2024; William Arruda, 2024).

However, while the integration of AI into SEO provides significant opportunities, it also brings challenges. Ethical concerns, such as algorithm transparency, data protection and potential biases, create significant barriers for marketers wishing to implement responsible and efficient strategies (Rajawat Manisha, 2024). The "black box" nature of AI systems often complicates accountability and decision-making, while the reliance on high-quality data raises questions about consumer consent and regulatory compliance (Akter et al., 2022; Lindebaum et al., 2020). In addition to ethical concerns, marketers increasingly express anxiety about AI's potential to replace human roles. There is growing evidence that the more professionals interact with AI tools, the more likely they are to fear being displaced by them. According to a 2023 CNBC/SurveyMonkey Workforce Survey, 60% of employees who use AI regularly worry about its impact on their job security, despite acknowledging its benefits for productivity (Jack Kelly, 2024).

1.2. Purpose and Research Questions

While the integration of AI into various areas of marketing and its impact on marketers' perceived efficiency and productivity has received considerable academic attention, empirical research specifically focused on the integration of AI into SEO remains limited (Haleem et al., 2022; Mohamed Riyath & Eid, 2025). SEO is a core element of digital marketing strategy, yet its transformation through AI technologies has not been extensively studied in academic literature. In addition, scientific literature directly addressing the challenges that SEO practitioners face with AI integration is limited. Nevertheless, the broader link between SEO and digital marketing points to several recurring challenges—such as the lack of transparency in AI algorithms, the dependence on high-quality data, and concerns over data privacy and ethics (Ziakis & Vlachopoulou, 2024). Another key concern is the fear of replacement due to automation, which can create psychological barriers that reduce motivation and hinder the adoption of AI tools (Mirbabaie et al., 2022). Moreover, while research highlights the role of AI knowledge in facilitating adoption and user engagement, its moderating effect on the successful integration of AI into SEO practices remains underexplored. This is particularly important given that only 37% of B2B marketers report effectively using AI, indicating a significant gap in AI literacy that could impact implementation outcomes (Moradi & Dass, 2022).

The aim of this master's thesis is to explore how the integration of AI is transforming SEO practices, with particular emphasis on its impact on marketers' perceived efficiency and productivity. At the same time, the study seeks to identify and analyze the key challenges that may hinder the effective implementation of AI in SEO, such as ethical concerns, fear of job replacement, and limited AI-related knowledge. By examining both the potential benefits and the barriers to adoption, the research aims to offer a comprehensive understanding of how